

Emotionally Intelligent Companion Dog Robot with AI-Driven Security and Voice Interaction

Sub Component (Face Recognition and Emotion Detection Subsystem for Security & Familiarity in Buddy Robot Dog)

R26-SE-040

Project Proposal Report

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March 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

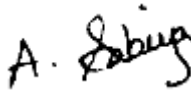
Name	Student ID	Signature
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Table 1

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

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Prof Pradeep Abeygunawardhana	14 th March 2026	

Table 2

ABSTRACT (Maximum 300 words)

Traditional home security systems mainly rely on cameras that record video footage but cannot automatically recognize individuals or understand the emotional state of users. As a result, homeowners must manually monitor surveillance systems, which is inefficient and time-consuming. In addition, these systems do not provide emotional interaction or companionship for users living alone.

This research proposes a **Face Recognition and Emotion Detection subsystem** for an emotionally intelligent robot dog named **Buddy**. The system enables the robot to identify familiar individuals, detect unknown persons, and analyze human emotional states in real time. The solution combines advanced computer vision and deep learning models including **MTCNN for face detection**, **FaceNet for facial embedding generation**, and **DeepFace for emotion recognition**. The entire processing pipeline operates on **edge hardware using a Raspberry Pi**, eliminating the need for cloud processing and improving user privacy.

The system performs two main functions. First, it acts as a **biometric security module** capable of distinguishing family members from strangers and sending alerts to a connected mobile application when unknown individuals are detected. Second, it acts as an **emotion monitoring companion**, continuously analyzing facial expressions to detect emotions such as happiness, sadness, anger, and neutrality.

An additional feature of the system is a **daily emotional monitoring algorithm** that calculates the percentage of emotional states detected throughout the day. If a user shows a prolonged negative emotional state such as sadness, the system triggers motivational responses through the robot's voice module to provide supportive interaction.

By combining **security monitoring with emotional awareness**, this research contributes to the development of affordable, privacy-preserving smart home robotics capable of improving both safety and human well-being.

Keywords: Face Recognition, Emotion Detection, Edge AI, Deep Learning, Smart Home Robotics

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my supervisor, **Prof. Pradeep Abeygunawardhana**, for his valuable guidance, encouragement, and support throughout the development of this research proposal.

I would also like to thank my project group members, **Ashandth Uthayashankar** and **SSY Wickramasinghe**, for their collaboration in designing and developing the integrated Buddy Robot system.

Finally, I would like to thank the **Sri Lanka Institute of Information Technology (SLIIT)** for providing the academic resources, facilities, and support required to undertake this final year research project.

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LIST OF ABBREVIATIONS

- AI: Artificial Intelligence
- CNN: Convolutional Neural Network
- MTCNN: Multi-Task Cascaded Convolutional Networks
- SDG: Sustainable Development Goals
- WBS: Work Breakdown Structure

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1. INTRODUCTION

1.1 Background and Context

With the rapid growth of smart home technologies, there is increasing demand for intelligent systems that can enhance both **home security and user well-being**. Traditional home security systems rely primarily on surveillance cameras and motion sensors. While these systems can record activity, they lack the ability to **identify individuals or analyze human emotions**, requiring users to manually monitor footage.

At the same time, modern society is experiencing an increase in **loneliness and mental stress**, particularly among individuals living alone or elderly populations. Technology has the potential to provide companionship and emotional support through interactive robotic systems.

Recent advancements in **Artificial Intelligence (AI)** and **deep learning** have made it possible for machines to analyze facial features and detect emotional expressions [8], [9]. Face recognition technology is widely used in smartphone authentication and surveillance systems, while emotion recognition is an emerging field within **affective computing**, which focuses on enabling machines to understand human emotions [10].

This project introduces **Buddy**, an emotionally intelligent robot dog designed to provide both **security monitoring and emotional companionship** within smart home environments. The proposed subsystem focuses on **face recognition and emotion detection**, enabling the robot to identify people and adapt its interactions based on their emotional states.

1.2 Literature Review

Face recognition technology has significantly improved with the introduction of deep learning models [9]. One of the most influential models is **FaceNet**, which converts facial images into numerical embeddings that represent unique facial characteristics [1]. These embeddings allow accurate comparison between different individuals using similarity measures.

Another widely used facial recognition framework is **DeepFace**, which integrates multiple deep learning models to achieve high recognition accuracy [2], [5]. DeepFace also includes modules for emotion recognition, age detection, and gender classification.

Emotion recognition systems typically rely on datasets such as **FER2013**, which contains thousands of labeled facial images representing different emotional expressions[4]. Deep learning models trained on these datasets can classify emotions such as happiness, sadness, anger, surprise, fear, disgust, and neutrality [8].

Face detection is a crucial first step in any facial analysis system. The **Multi-Task Cascaded Convolutional Network (MTCNN)** is commonly used for face detection because of its high accuracy in detecting faces under different lighting conditions, orientations, and scales [3].

Although these technologies are widely studied individually, most implementations are limited to either **security systems or standalone emotion detection applications**. Very few systems combine **real-time face recognition, emotion analysis, and robotic interaction in a low-cost edge computing environment**.

1.3 Research Gap

Despite the progress in computer vision and robotics, several limitations remain in current systems.

Most home security solutions focus only on **detecting motion or recording video**, without identifying individuals. In addition, many existing facial recognition systems rely heavily on **cloud computing**, raising privacy concerns because sensitive biometric data must be transmitted to remote servers.

Emotion recognition systems are also commonly developed as **independent research prototypes**, without integration into real-world applications such as autonomous robots.

Furthermore, commercially available robotic companions such as **Sony Aibo** are extremely expensive and inaccessible to most households.

Therefore, there is a clear need for a **low-cost robotic system capable of performing face recognition and emotion analysis locally using edge computing**, while also providing interactive responses that support user well-being.

2. OBJECTIVE

2.1 Main Objective

To develop a **real-time face recognition and emotion detection system** that enables the Buddy robot dog to identify individuals, detect strangers, and analyze emotional states in order to provide both security monitoring and emotionally aware interactions.

2.2 Specific Objectives

- Implement an accurate face detection mechanism using the MTCNN deep learning model.
- Generate unique 128-dimensional face embeddings using the FaceNet model for biometric identification.
- Develop a **face recognition system using cosine similarity** to distinguish between known individuals and unknown persons.
- Implement emotion detection using DeepFace models trained on the FER2013 dataset.
- Design an **emotion monitoring algorithm** that calculates daily emotional state percentages.
- Integrate the subsystem with the robot's **voice and movement modules** to generate appropriate emotional responses.
- Implement **real-time alert notifications** when unknown individuals are detected.

3. METHODOLOGY

The proposed system follows a computer vision pipeline designed for **real-time facial analysis on edge hardware**.

3.1 Face Detection

The first stage of the system detects faces within the camera feed using the **Multi-Task Cascaded Convolutional Network (MTCNN)** [3]. This model detects facial bounding boxes and key landmarks such as the eyes, nose, and mouth.

Detected faces are then extracted and resized to the required input size for further processing.

3.2 Face Recognition

After face detection, the system generates facial embeddings using the **FaceNet deep learning model** [1]. FaceNet converts each face image into a **128-dimensional numerical vector** representing unique facial features.

These embeddings are stored in a local database. When a new face is detected, its embedding is compared with stored embeddings using **cosine similarity**. If the similarity exceeds a predefined threshold, the person is recognized as a known individual.

3.3 Emotion Detection

Emotion detection is implemented using **DeepFace** [5], which uses convolutional neural networks trained on the **FER2013 dataset** [4]. The model classifies facial expressions into seven emotional categories:

- Happy
- Sad
- Angry
- Fear
- Surprise
- Disgust
- Neutral

Each prediction includes a confidence score representing the probability of each emotion.

3.4 Emotion Monitoring Algorithm

The system continuously logs detected emotions throughout the day. A statistical algorithm calculates the **percentage of time each emotional state occurs**.

If the system detects that a user shows a high percentage of negative emotions, such as prolonged sadness, the robot triggers motivational interactions through its voice module.

3.5 System Implementation

The system is implemented using the following technologies:

Hardware:

- Raspberry Pi 4 Model B
- Raspberry Pi Camera Module

- Servo motor control system
- Ultrasonic sensors

Software:

- Python
- OpenCV [7]
- DeepFace [5]
- TensorFlow [6]
- FaceNet [1]

3.6 Evaluation Metrics

System performance will be evaluated using the following metrics:

- Face recognition accuracy
- False Acceptance Rate (FAR)
- False Rejection Rate (FRR)
- Emotion classification accuracy
- Processing speed (frames per second)

4. HIGH-LEVEL SYSTEM ARCHITECTURE

System Flow: Camera → Face Detection (MTCNN) → Face Alignment → Face Embedding Generation (FaceNet) → Identity Matching → Emotion Detection (DeepFace) → Robot Response

Hardware & Software: The system utilizes a camera module and a Raspberry Pi 4. The software stack includes distinct models for face detection, embedding, emotion classification, and identity matching.

- **System Outputs:**
 - **Stranger Detected:** Captures an image and sends an alert notification directly to the mobile app.
 - **Known Person:** The robot greets the user and adjusts behavior based on current emotion (e.g., energetic movement if happy, gentle interaction if sad).
- The system begins by capturing real-time video frames through the Raspberry Pi camera module. Each frame is processed to detect faces, which are then aligned and converted into embeddings using FaceNet.
- The generated embeddings are compared with stored embeddings to identify individuals. Simultaneously, the detected face is analyzed by the emotion detection module to determine the user's emotional state.
- Based on these results, the robot performs appropriate actions such as greeting the user, sending alerts for unknown individuals, or providing emotional support when needed.

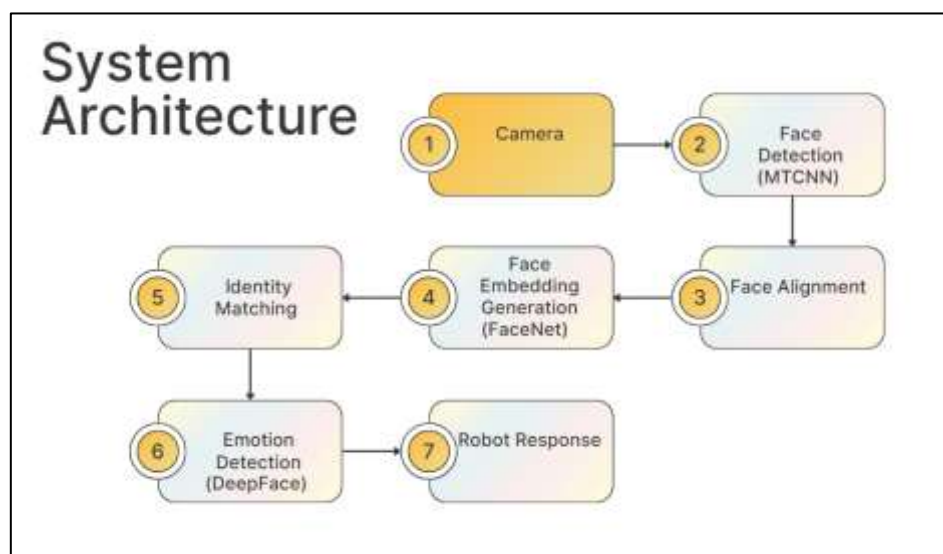


Figure 1

5. USER REQUIREMENTS

Stakeholders

- Homeowners
- Smart home users
- Elderly individuals living alone
- Families seeking enhanced home security

Functional Requirements

- The system must detect human faces in real time.
- The system must identify known individuals using facial recognition.
- The system must detect unknown individuals and trigger alerts.
- The system must analyze facial expressions to determine emotional states.
- The system must track emotional patterns throughout the day.

Non-Functional Requirements

Privacy

The system must store only facial embeddings locally, avoiding storage of raw images.

Performance

The system must operate in real time on edge hardware.

Security

All biometric data must be stored securely to prevent unauthorized access.

Scalability

The system must allow new users to be added easily without retraining the entire model.

6. COMMERCIALIZATION PLAN

The Buddy robot dog has potential applications in **smart homes, security monitoring, and emotional companionship systems.**

Target Market

- Smart home users
- Elderly care facilities
- Small security companies
- Educational robotics institutions

Revenue Model

Revenue may be generated through:

- Hardware sales
- Software updates and upgrades
- Mobile application subscriptions

Competitive Advantage

Unlike traditional home security systems, the Buddy robot combines:

- Face recognition
- Emotion detection
- Interactive companionship

into a single affordable robotic platform.

7. BUDGET AND JUSTIFICATION

The Buddy Dog Robot prototype is developed with an approximate budget of LKR 60,000.

Component	Estimated Cost (LKR)	Justification
Raspberry Pi 4	18,000	Required for edge computing and running offline AI models.
Arduino Uno	5,000	Microcontroller for sensor integration.
MG996R Servo Motor (8–10 motors)	20,000	Actuation for the robot's limbs.
HC-SR04 Ultrasonic Sensor	1,500	Required for distance calculation and obstacle avoidance.
MPU6050 Sensor	1,200	Required for balance and movement control.
Raspberry Pi Camera Module	5,000	Required for visual input and face recognition integration.
PCA9685 Servo Driver	2,500	Required to drive multiple servo motors simultaneously.
Battery and Power System	4,000	Power supply for mobile autonomy.
Frame and Mechanical Parts	2,800	Chassis and physical structure.
Total	~60,000	

Table 3

8. WORK BREAKDOWN STRUCTURE (WBS)

The project will be completed through the following stages:

1. Literature review and dataset collection
2. Face detection model implementation
3. Face recognition model development
4. Emotion detection model integration
5. Hardware assembly and integration
6. System testing and performance optimization
7. Final evaluation and documentation

9. GANTT CHART

Phase	Duration
Research and system design	Month 1–2
AI model development	Month 3–4
Hardware integration	Month 5–6
Testing and optimization	Month 7–8
Final evaluation and documentation	Month 9

Table 4



Figure 2

10.REFERENCES LIST

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11.APPENDICES

- Appendix A – Privacy protocol for storing facial embeddings
- Appendix B – Emotion classification mapping
- Appendix C – System integration diagrams